Extensions of Positive Linear Functionals and the N-Representability Problem

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One of the oldest unsolved problems in Quantum Chemistry is the "The N-Representability Problem" which is to determine whether a positive matrix that acts on the Hilbert space of two electron states is in fact derived from an N-electron density matrix and thus is a Second Order Reduced Density Matrix (SORDM). If one had an *intrinsic* solution of this problem, i.e. a solution that only involved SORDM matrix elements, then one could reduce the number of variables needed to variationally determine the N-electron ground state from being proportional to r^N (obviously dependent on the number of electrons in the system) to being proportional to r^4,(not dependent on the number of electrons in the system) where r is the dimension of the approximating one electron Hilbert space.

On this 50^{th} anniversary of the first Sanibel conference it seems appropriate and instructive to review the basic formulation of the N-Representability problem and scan (incompletely) some of the important milestones achieved on the yet incomplete journey to a solution. At the same time I will recast the problem in terms of extensions of positive linear functionals defined on subspaces of Fock space. The full N-electron density matrix determines a bounded positive linear functional on the vector spaces of N-electron operators, while SORDM's define positive linear functionals on the subspace of 2-electron operators. In this context the N-Representability problem becomes one of determining the conditions a positive linear functional defined on a subspace must satisfy in order to be extended to a positive linear functional defined on a space containing this subspace.

In this presentation I will review some results from the mathematical literature and apply them to the N-Representability problem for First Order Reduced Density Matrices (FORDM's) (a solved problem) and for SORDM's leading to *Necessary and Sufficient intrinsic* conditions for such extensions to exist. The problem of how to apply these conditions in a meaningful and practical way still remains elusive but viewing the N-Representability problem from this perspective hopefully will enhance the prospects of finding a practical solution.